Team 2

# Customer Insights and Data Analytics Early Remote Internship

## Project Title: Optimizing Customer Support Performance Through Strategic Ticket Analytics

Presented By


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# Our 4 week Internship's journey

**WEEK-1**

**Ticket Mapping & Recurring Issue Analysis**

Mapped tickets, identified recurring issues, and designed support flows.

**WEEK-2**

**Data Cleaning & Tagging**

Cleaned data, applied tagging and sentiment, and extracted priority insights.

**WEEK-3**

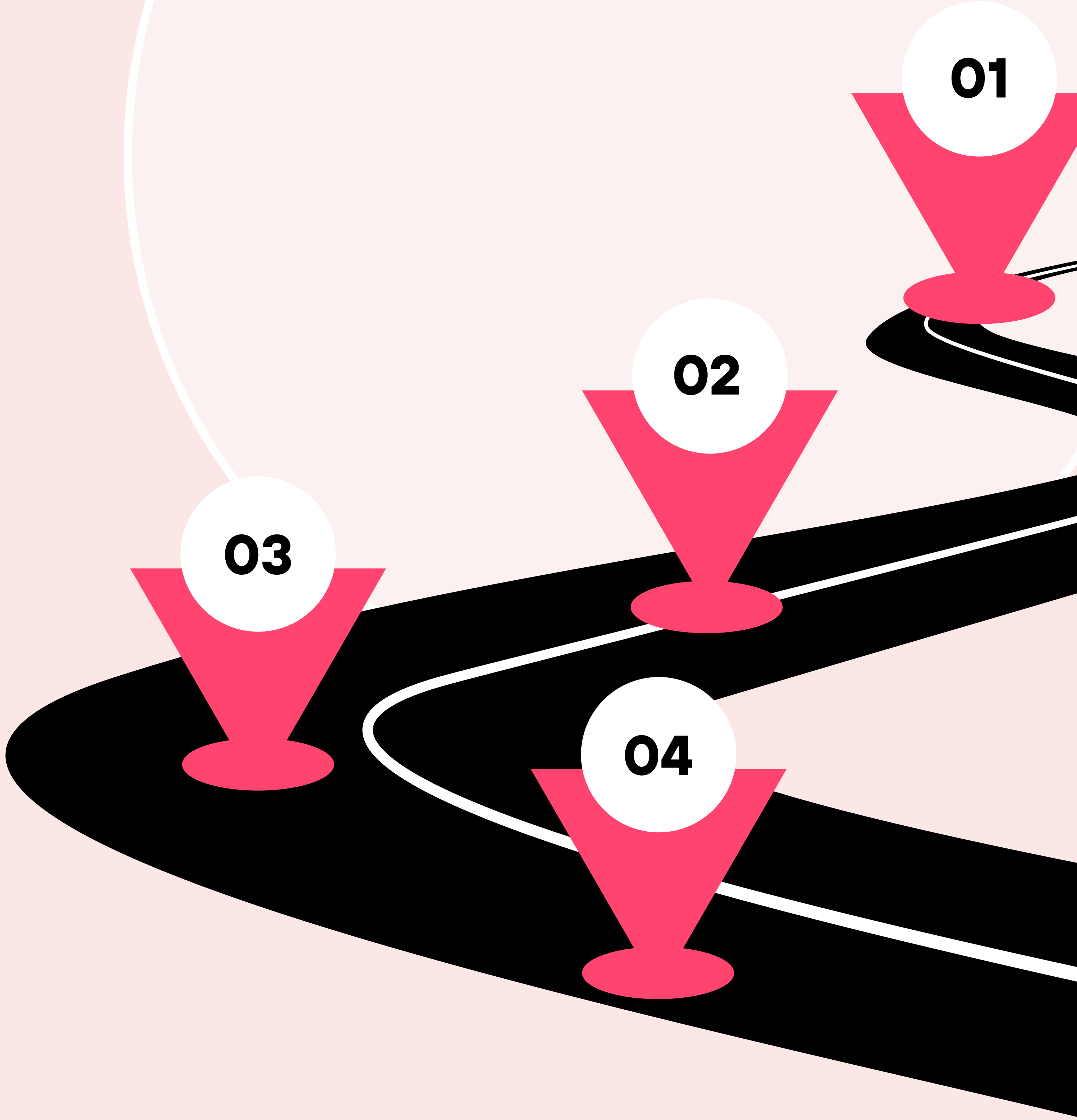
**Dashboard Insights & Forecasting**

Built Power BI dashboards, found bottlenecks, and forecasted volumes.

**WEEK-4**

**Strategic Recommendations & Reporting**

Linked insights to CSAT, proposed automation/ triage/ self-service, and delivered the final report and presentation.



# Dataset Overview

The dataset contains customer support ticket details for analysis of service performance.

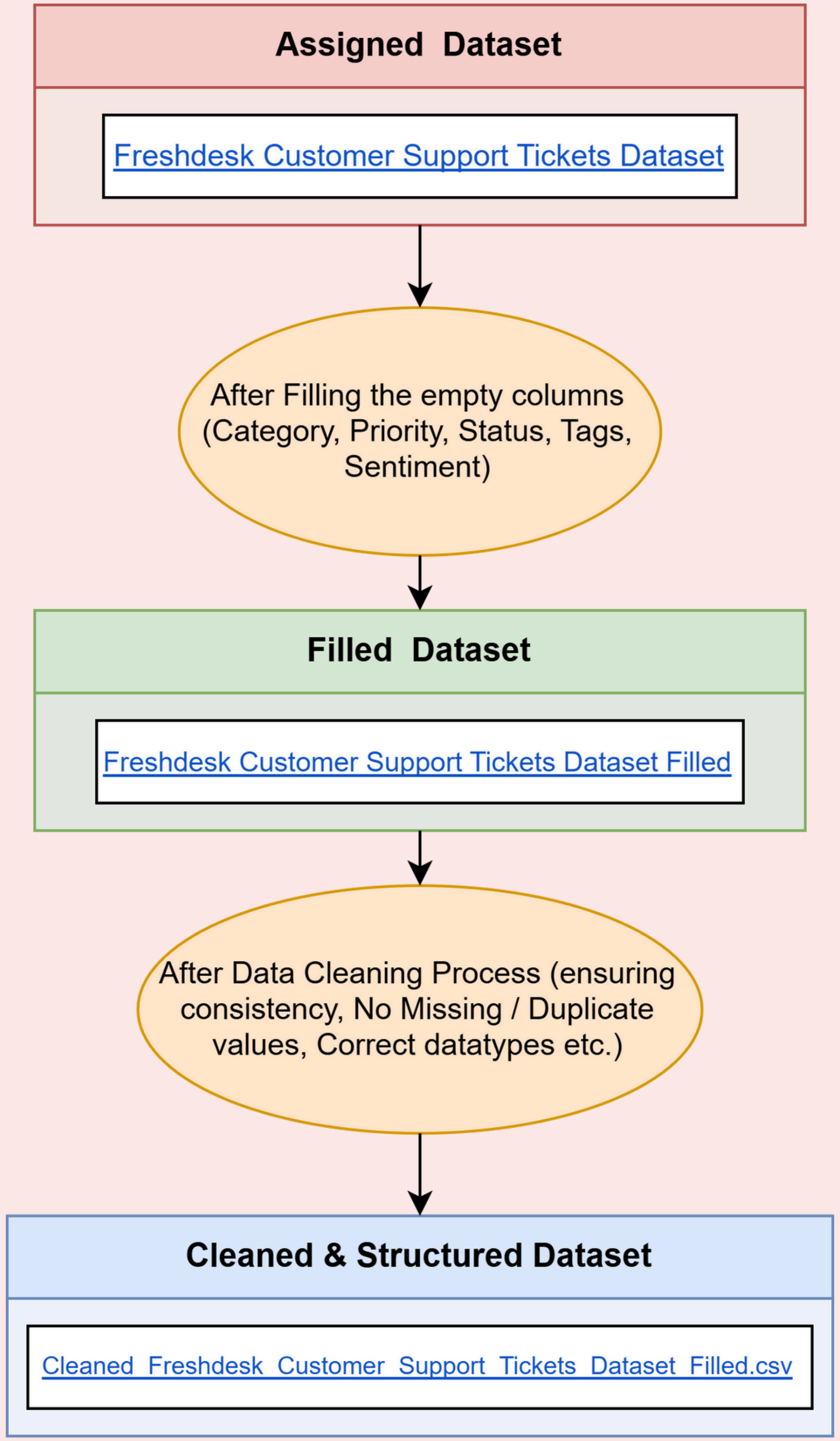
Key columns include **Ticket ID**, **Created Date**, **Resolved Date**, **Category**, **Priority**, **Status**, **Agent**, **Tags**, **Sentiment**, and **Subject/Short Description**.

***Time-based fields*** *like **Created Date** and **Resolved Date** enable trend analysis and resolution tracking.*

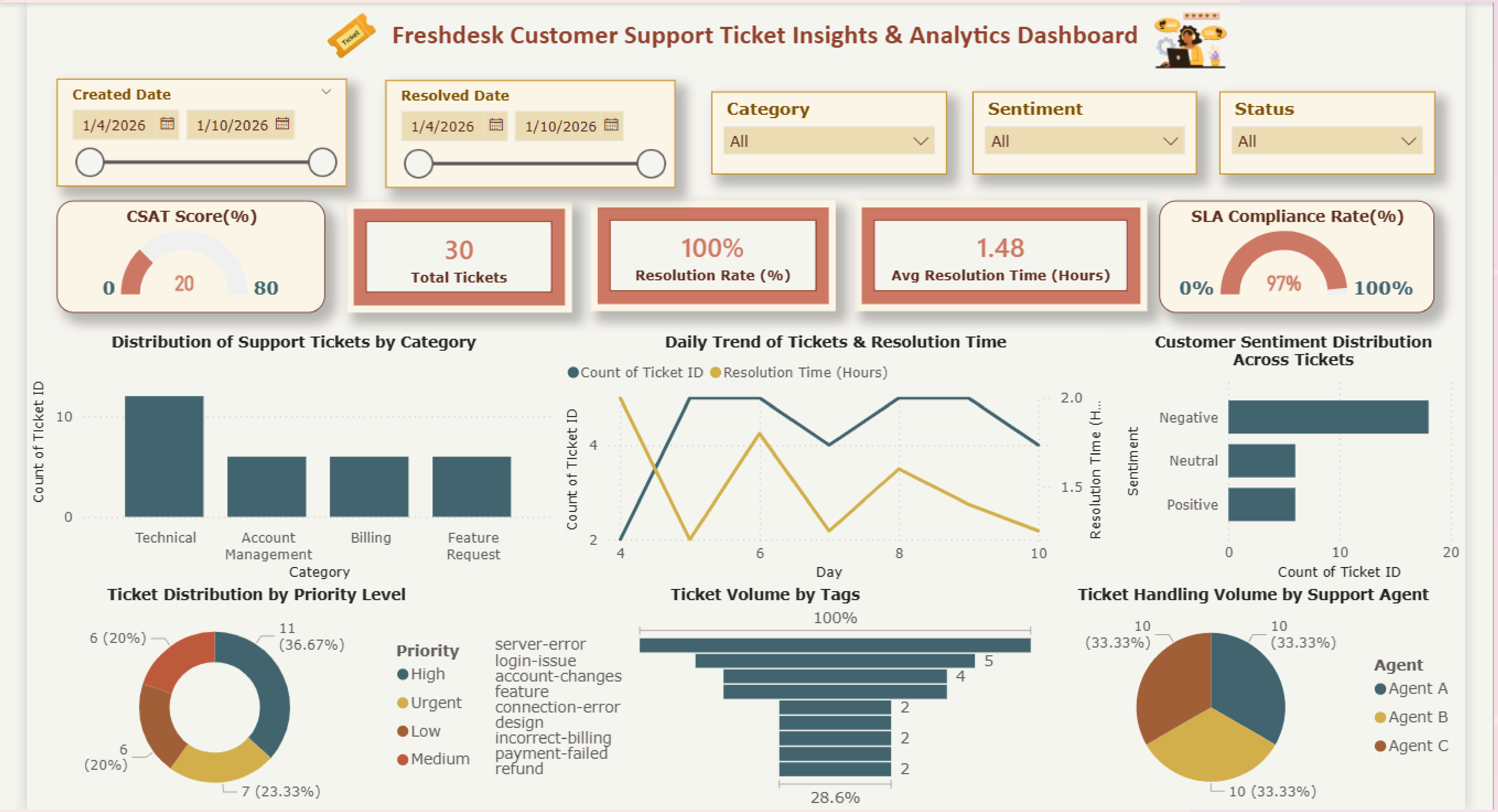
***Categorical fields*** *such as **Category**, **Priority**, **Status**, **Agent**, and **Tags** support identification of recurring issues and workload patterns.*

**Experience-related fields** like **Sentiment** reflect customer satisfaction, helping link operational metrics with service quality.

Overall, these structured date/time, categorical, and text columns allow detailed analysis of support operations, bottlenecks, and customer experience trends.



# Freshdesk Customer Support Ticket Insights & Analytics – Power BI Dashboard




## KPIs

- 30 tickets show strong operational efficiency but weak customer experience
- 100% resolution rate, 1.48-hour avg resolution, 97% SLA → high process performance
- CSAT 20% → very low customer satisfaction
- Reveals a speed-focused model where tickets close fast but problems remain unresolved

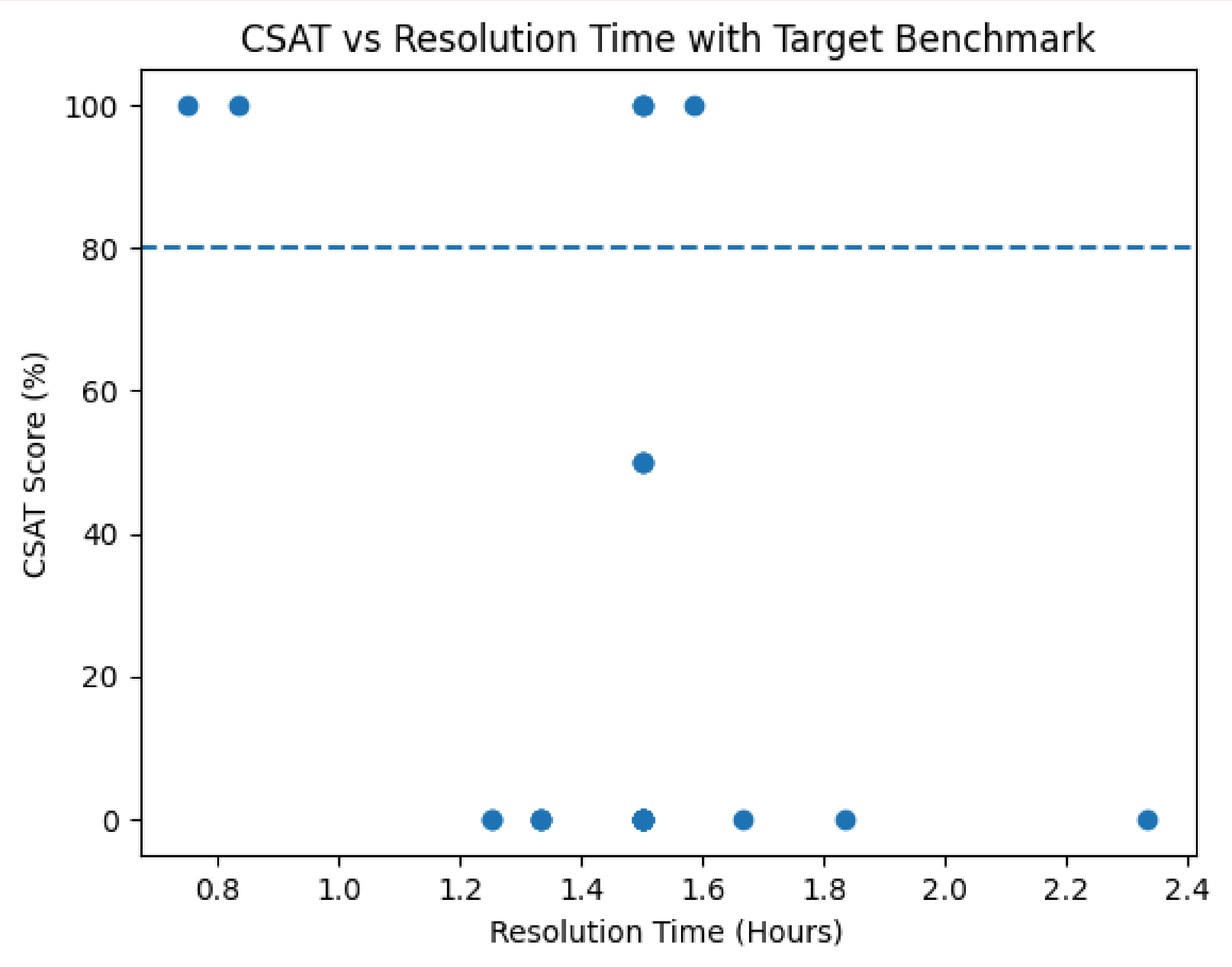
## Filters

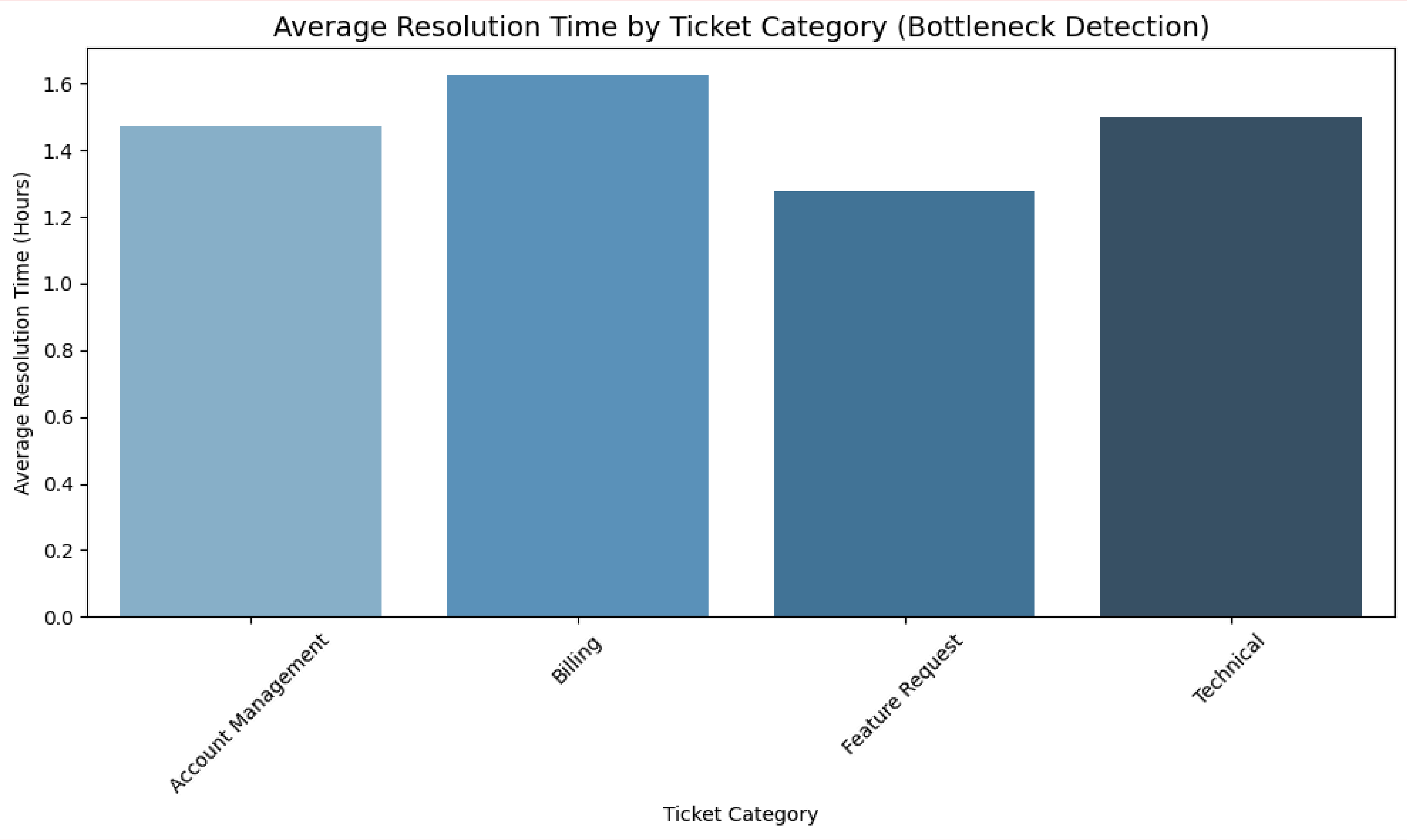
- Filters by date, category, sentiment, and status enable targeted analysis
- Enable targeted issue analysis
- Identify urgent cases, technical patterns, and trends
- Support root-cause detection

## Visualizations

- Fast closures & balanced workloads → strong operational efficiency
- High technical tickets & server/login issues → product quality bottlenecks
- High urgent-priority volume → escalation pressure
- Negative customer sentiment → poor service experience
- Trend spikes & delayed urgent tickets → triage and escalation failures
- Speed-focused operations → weak alignment with customer satisfaction & issue severity

# Advanced  Additional Analytics Insights from Support Operations



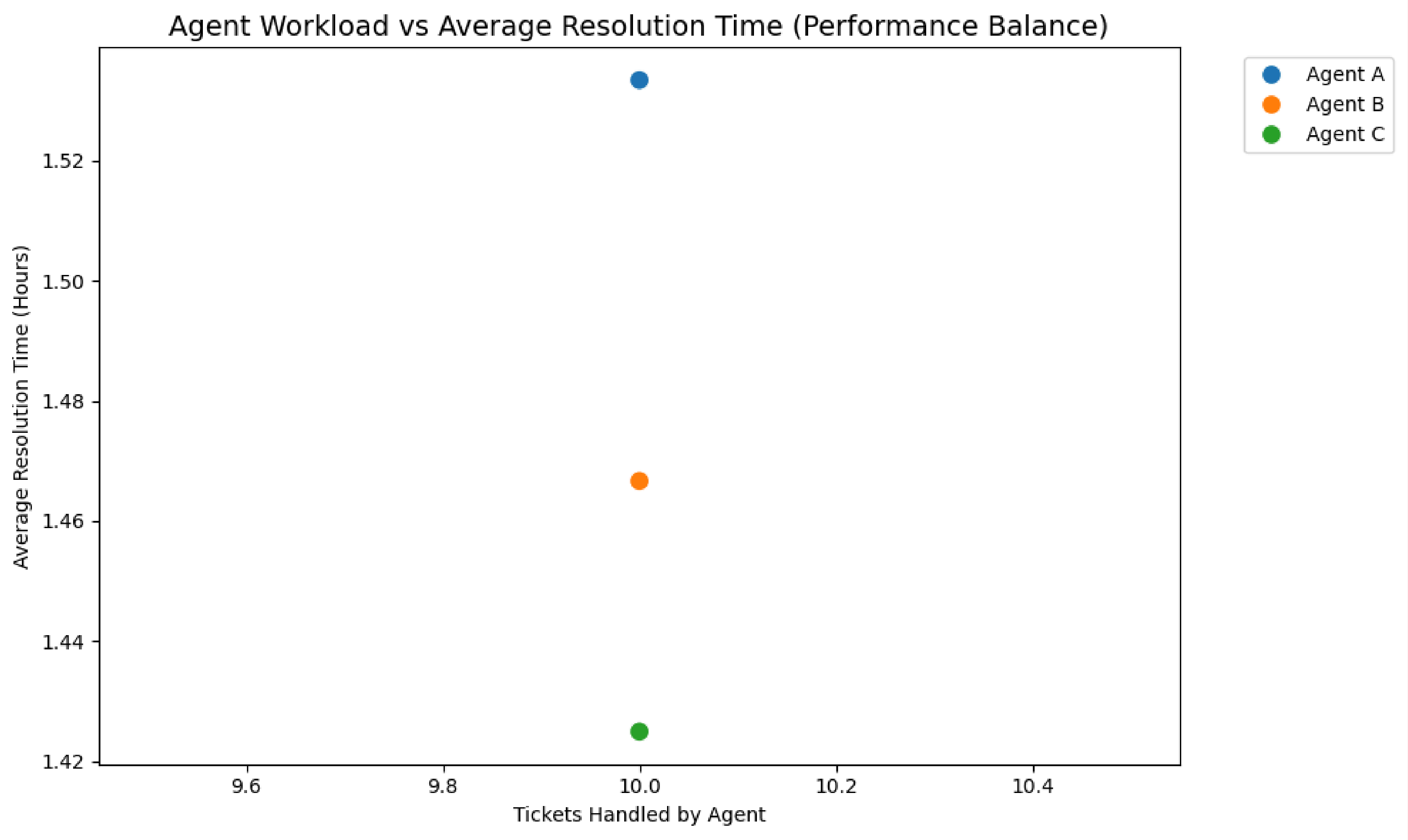

## CSAT vs Resolution Time (Scatter Plot)

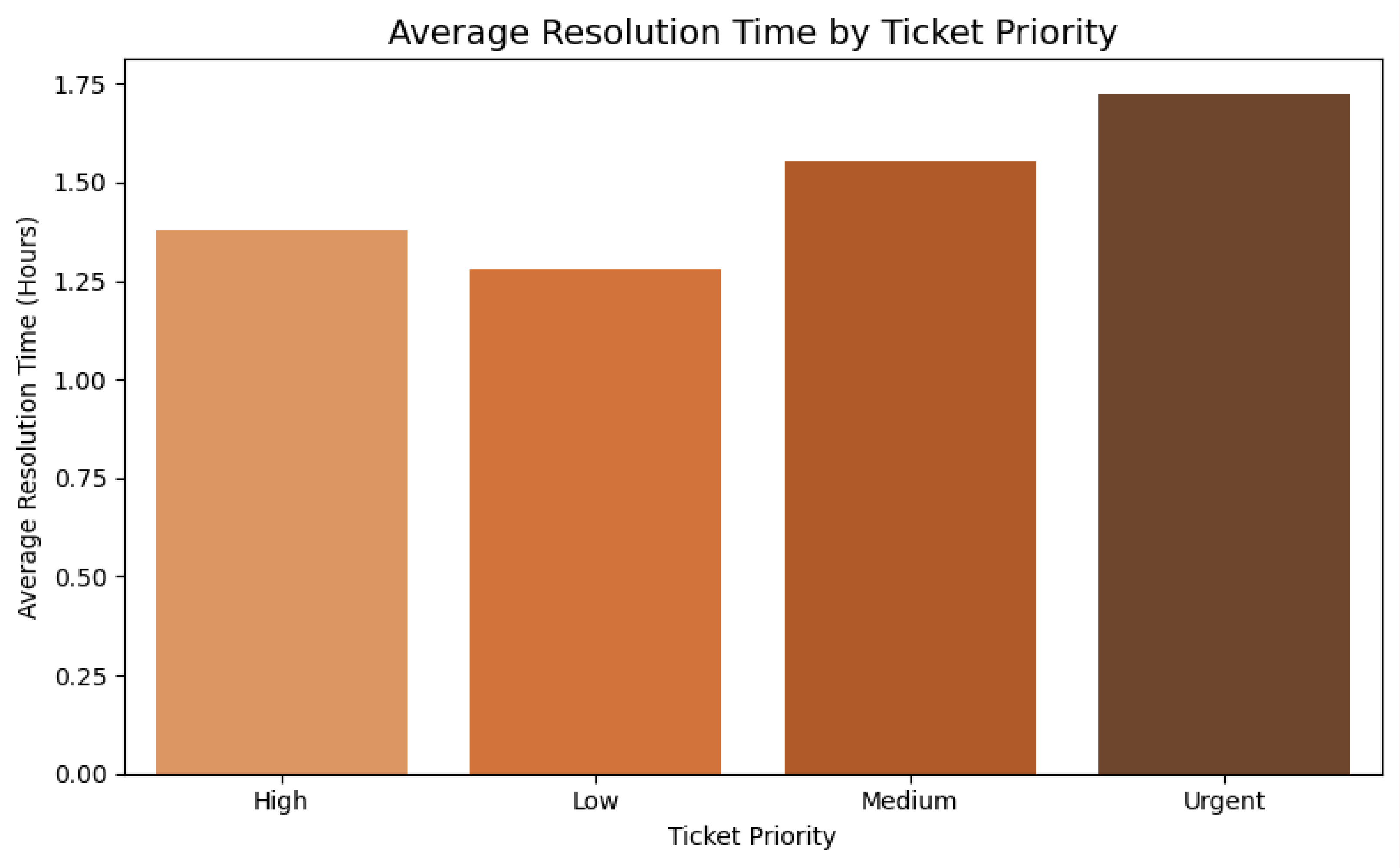
Shows the relationship between speed and satisfaction, revealing that faster resolution generally leads to higher CSAT, but not always. Some fast closures still result in low satisfaction, proving that **quality of resolution matters as much as speed**. Highlights the need to balance quick response with effective problem-solving.

## Category vs Resolution Time (Bottleneck Detection – Bar Chart)

Identifies **Billing and Technical** categories as major bottlenecks due to longer resolution times. Indicates process complexity and resource gaps in high-impact areas. Points to the need for **specialized handling, process optimization, and better resource allocation**.

# Advanced  Additional Analytics Insights from Support Operations



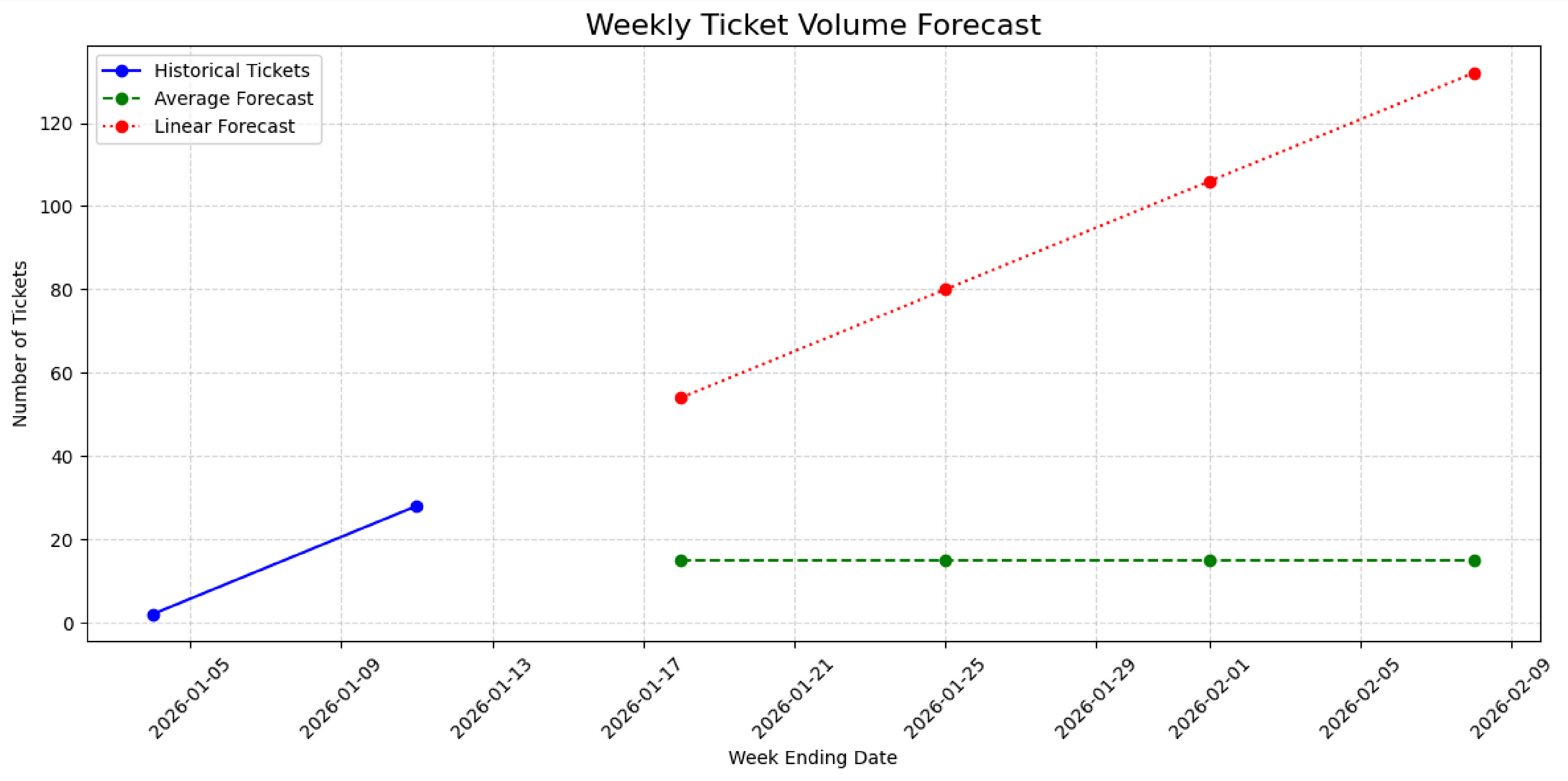

## Agent Workload vs Resolution Time (Performance Balance – Scatter Plot)

Reveals performance differences despite equal workload distribution. Shows that **uniform ticket allocation hides efficiency gaps** and limits productivity. Supports the need for **skill-based routing and targeted training** instead of rigid round-robin assignment.

## Priority vs Resolution Time (Bar Chart)

Exposes **priority inversion**, where urgent tickets take longer than low-priority ones. Indicates weaknesses in **triage, escalation, and workflow prioritization**. Emphasizes the need for **stronger priority handling protocols** to protect critical customer issues.

# Ticket Volume Forecast & Insights




**Purpose:** Estimate future ticket volumes to plan staffing, resources, and operations effectively.

**Forecast Approaches:**

- **Weekly Average Forecast:** Stable volume based on historical weekly average.
- **Linear Trend Forecast:** Growth-oriented projection using historical trend (linear regression).

**Key Observations:**

**Historical:** Ticket volume surged early Jan 2026 (2 → 28).

**Average forecast:** ~15 tickets/week → conservative planning. (stable)

**Linear forecast:** projected rapid growth (54 → 132) → highlights potential peak periods. (growth trend)

🎯 Trend divergence emphasizes need for proactive vs. routine planning.

**Operational Insight:** Using both forecasts helps balance daily staffing and prepare for high-demand periods.

# Strategic Recommendations for Workflow Improvements

## CSAT Score Improvement — 01

- Use post-resolution survey instead of sentiment
- 5-point rating + short feedback
- Track satisfaction accurately

## Recurring Technical Failures — 02

- Regular server tests and bug fixes
- Strengthen tech team & maintenance
- Reduce system-driven tickets

## Billing Delays & Complexity — 03

- Assign specialized billing agents
- Expand billing team
- Improve speed & customer trust

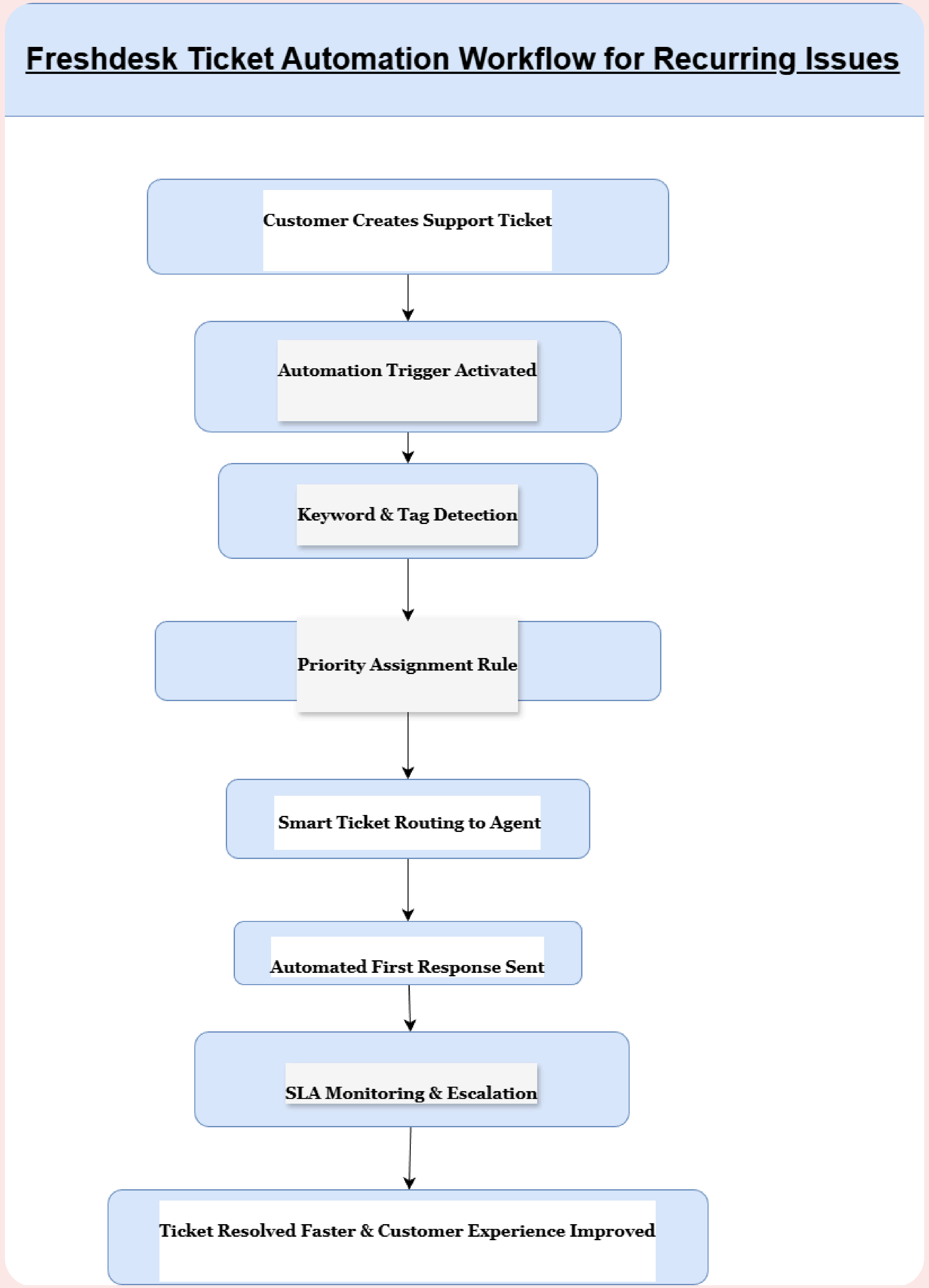
## Unstrategic Ticket Handling — 04

- Priority-based routing & deadlines
- Assign agents by skill
- Resolve urgent issues faster

## Agent Burnout Risk Reduction — 05

- Use automation, chatbot, FAQs, self-service
- Cut repetitive tickets
- Balance workload & boost CSAT

# Freshdesk Scenario – Automation Design




**Case: New Ticket Created**

**Step 1: Auto-Tagging & Category Detection**

- Scans subject, category, tags automatically
- Keywords (login, server, payment) trigger tagging
- Speeds up routing, reduces manual errors

**Step 2: Priority Assignment Rule**

- High-impact issues → High/Urgent
- Low-impact issues → Medium/Low
- Ensures critical tickets handled first

**Step 3: Smart Ticket Routing to Specialized Agents**

- Technical → Tech Support Agent
- Billing/Refund → Finance Agent
- Balances workload, leverages expertise

**Step 4: Automated Customer Response (Macro Reply)**

- Instant acknowledgment for recurring issues
- Includes troubleshooting or ETA
- Keeps customers informed

**Step 5: SLA-Based Escalation Trigger**

- Unresolved high-priority tickets → senior agent
- Prevents SLA breaches, ensures timely resolution

# Our Acknowledgement & Conclusion

### Key Learnings from the Internship:

We gained hands-on experience in ticket analysis, data cleaning, and issue tagging, built insights using Power BI dashboards, KPIs, and visualizations, and designed actionable workflow improvements through automation and priority alignment. As a team, we developed end-to-end skills in transforming raw support data into strategic, data-driven recommendations that improve both operational efficiency and customer satisfaction.


### Acknowledgement:

We sincerely thank our mentors, the internship team, and teammates for their guidance, feedback, and support throughout this learning journey.


**Our Note: "Effective customer support is built on insight, action, and empathy—every ticket is an opportunity to improve and delight the customer"**

**Our Virtual Presentation Recording (link):**
https://drive.google.com/file/d/16FUkx-Z3x2kQfzOvdgWyMZNUXlVotYwu/view?usp=sharing

# Thank You

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